

Direct and indirect uncertainties in the prediction of tilted irradiance for solar engineering applications

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Abstract

Global radiation measured on fixed-tilt, south-facing planes (40° and vertical) and a 2-axis tracker at NREL's Solar Radiation Research Lab. in Golden, CO is compared to predictions from ten transposition models, in combination with either optimal or suboptimal input data of horizontal irradiance. Suboptimal inputs are typically used in everyday engineering calculations, for which the necessary data are usually unavailable for the site under scrutiny, and must be estimated in some way. The performance of the transposition models is first evaluated for ideal conditions when optimal data are used. In this specific case, it is found that the Gueymard and Perez models provide the best estimates of global tilted irradiance under clear skies in particular.

The performance of four direct/diffuse separation models is also evaluated. Their predictions of direct and diffuse radiation appear biased in most cases, with a model-dependent magnitude. Finally, the performance of the resulting combinations of separation and transposition models is analyzed in a variety of situations. When only global irradiance is known, the accuracy of the tilted irradiance predictions degrades significantly, and is mainly conditioned by the local performance of the direct/diffuse separation method. For the south-facing vertical surface, inaccuracies in the ground reflection calculations becomes another key factor and significantly increase the prediction error. The Reindl transposition algorithm appears to perform best in this case. When using suboptimal input data for the prediction of plane-of-array irradiance on a moderately tilted plane (40°S) or a 2-axis tracking plane, the Hay, Reindl and Skartveit models are less penalized than others and tend to perform better. It is concluded that further research should be conducted to improve the overall process of predicting irradiance on tilted planes in realistic situations where no local high-quality irradiance or albedo measurements are available.

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1. Introduction

Accurate solar radiation resource data are necessary at various steps of the design, simulation, and performance evaluation of any project involving solar energy. Solar energy systems are installed on either fixed tilted planes or tracking receivers. Similarly, glazed envelopes are installed vertically on the facades of buildings or at a vari-

ety of tilt angles in atria. Prediction of the irradiance incident on such tilted surfaces is key to the evaluation of the effective solar resource and of the performance of all these systems. Because of the usual lack of measured data at the project's site, the solar resource needs to be modeled in most cases. Various models can be used, depending on the time resolution (e.g., hourly or monthly mean) and on the type of available input data (e.g., meteorological information or global irradiance on the horizontal). A typology of such models has been proposed recently (Gueymard and Myers, 2008a). Monthly and hourly datasets of global radiation on the horizontal can now be found from

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various sources, such as NASA's SSE (Stackhouse et al., 2006), NREL's NSRDB (NREL, 1995) and SWERA (René et al., 2005). These and other sources of measured or modeled data are described elsewhere (Gueymard and Myers, 2008b).

Before the prediction of tilted (or “plane-of-array”) irradiance can be attempted in practice, a first series of models is generally needed to predict the global irradiance on the horizontal and the direct irradiance on a plane perpendicular to the sun rays, or direct normal irradiance (DNI). If DNI predictions are accurate, no further calculation is needed for applications involving double-axis tracking concentrators with small acceptance angles. More generally, modeling the direct irradiance incident on a tilted surface is trivial if DNI is known, since only a geometric relationship is involved. Assessing the prediction performance of global irradiance or DNI with the various types of models currently available is beyond the scope of this contribution, although their accuracy indeed conditions that of the tilted irradiance. This issue is actually central to the discussion about the effect of using “suboptimal” inputs that is offered in Sections 5 and 6.

For most other applications involving flat-plate collectors or buildings, another difficulty is to estimate the diffuse irradiance that emanates from the sky and foreground reflections. All models reviewed here ideally assume that this foreground is horizontal and extends infinitely in front of the receiver with no interfering mask or shading surfaces. Realistic installations usually involve more geometrical and solar radiation calculation complexity, but such an analysis is beyond the scope of this contribution.

Up to now, most solar power projects were “small scale” and were not hurt much by solar resource miscalculations, thanks to error cancelations through the various calculation steps or engineering tolerances. Due to the extremely rapid expansion of the solar power plant industry, things are different now, and long-term errors as low as $\pm 5\%$ may jeopardize the profitability of large-scale projects, which is of serious concern since they require considerable investments. Various radiation datasets or calculation methods exist but they often disagree substantially (Myers, 2007; Suri et al., 2008). This is highly confusing and detrimental to the whole community. Furthermore, the current trend in the energy production field (particularly when using photovoltaic technologies) is to use realistic energy simulations of a solar power plant that are based on radiation data measured at high frequency, with time steps of one minute or less. Most transposition models have been developed, and so far tested, using hourly data. One of the objectives of the present study is to evaluate whether such models can be used reliably for 1-minute predictions. Many solar power plants now use flat photovoltaic panels. To maximize energy production, these flat-panels are frequently mounted on 2-axis trackers. Hence the emphasis added on the results obtained for such a configuration, which has been rarely studied in the literature.

A previous study (Gueymard and Myers, 2009) showed that the accuracy of the predicted global tilted irradiance was a function of the experimental error in the global horizontal irradiance and of other factors, including ground albedo and frequency distribution of the diffuse/global ratio. The goal of the present contribution is to generalize these results by considering more variables, and to obtain results that can be related to the everyday practice of solar energy engineering. In this study, the overall performance of the radiation transposition between the horizontal and tilted planes is assessed in various ways. The literature is rich in investigations of this kind (e.g. Hay and McKay, 1986; Kambezidis et al., 1994; Loutzenhiser et al., 2007; Psiloglou et al., 1996). These contributions, however, focus on the performance of models that evaluate the tilted diffuse irradiance under *ideal* conditions only, i.e., when both the direct and diffuse horizontal components are measured at first-class experimental sites, as well as other important ancillary data, such as ground albedo. These studies therefore fail to address the additional uncertainties introduced by the models used to separate the direct and diffuse horizontal components of global irradiance, or (*a fortiori*) those used to evaluate the global irradiance from meteorological inputs (such as cloud cover, aerosol turbidity, and water vapor).

In this respect, the present contribution offers some important improvements over the conventional validation and performance assessment methodologies that have been used in the literature up to now. These improvements include the use of:

- One-minute radiation data for enhanced time resolution, of importance for fast-responding (e.g., PV) energy systems.
- Measured (each minute) *vs* estimated (using a fixed value, e.g., 0.2) ground albedo, for snow-free *vs* snow-covered ground conditions.
- Accurate 1-minute cloud cover information.

These improvements provide the necessary elements to evaluate the uncertainty in the resulting global tilted irradiance¹ under various conditions, from ideal or “optimal” (when the input direct and diffuse irradiances, and the ground albedo, are known very accurately) to most usual or “suboptimal” (when only the global horizontal irradiance is available). It is emphasized that the main purpose of this study is not to compare the performance of various transposition algorithms at a single site (which would be

¹ The sum of direct and diffuse irradiance is usually referred to as “global irradiance”. The alternate term “total irradiance” is not used here because of its ambiguity (it is also frequently considered as synonymous of “broadband irradiance”, as opposed to “spectral irradiance”). Another alternative is “hemispherical irradiance”, as recommended by some standardization institutions (see, e.g., ASTM, 2003), but it is rarely used in the literature.

essentially of local interest), but to provide better understanding of what drives the modeling errors in general, and what experimental difficulties are involved in model validation, all of which is of universal interest. Here, the main difference with the existing literature is that a broader perspective is adopted to evaluate the prediction accuracies of global tilted irradiance over a more complete chain of calculations, thus closing part of the gap between scientific abstraction and practical engineering situations.

2. Transposition and separation models

The global irradiance, E_s , on a tilted plane, whose tilt is s degrees from the horizontal, can be evaluated from the classic equation:

$$E_s = E_{bn} \cos \theta + E_d R_d + \rho E R_r \quad (1)$$

where E_{bn} is DNI, E_d is the diffuse horizontal irradiance, E is the global horizontal irradiance, θ is the angle of incidence of the sun rays on the tilted plane, R_d is the diffuse transposition factor, ρ is the foreground's albedo, and R_r is the transposition factor for ground reflection. The irradiance components that appear on the right-hand-side of Eq. (1) are inter-related by the fundamental closure equation

$$E = E_{bn} \cos Z + E_d \quad (2)$$

where Z is the sun's zenith angle at any instant.

The calculation of the tilted direct component, $E_{bn} \cos \theta$, is purely geometric, and therefore straightforward and identical in all cases considered in what follows.

The evaluation of the ground-reflected diffuse irradiance is dependent on the factor R_r . Most studies consider that the ground reflection process is ideally isotropic, in which specific case R_r can be simplified into

$$R_r = (1 - \cos s)/2. \quad (3)$$

The foreground albedo can be estimated in different ways: *a priori* (as in the general case), experimentally measured on-site (as in this study), or modeled (Gueymard, 1987; Loutzenhiser et al., 2007). The uncertainties associated with both the use of approximate values of ρ and the isotropic approximation for R_r are discussed in Section 4.2.

The main unknown in Eq. (1) is normally R_d . If the diffuse radiance were ideally constant over the whole sky hemisphere, R_d would be easily obtained from the simple isotropic approximation

$$R_d = (1 + \cos s)/2. \quad (4)$$

In reality, a plane of tilt s facing the sun receives more diffuse radiation than a plane of same tilt in the opposite direction. Elaborate transposition models have been developed to estimate this anisotropic effect and calculate a refined value of R_d .

The transposition models selected here are those that are commonly used in solar energy engineering practice, for applications such as PV design (Carr, 2005) or building

energy simulation (Loutzenhiser et al., 2007). Besides the conventional isotropic approximation of Eq. (4), various anisotropic diffuse models have been proposed in the literature since the pioneering work of Threlkeld (1962). Threlkeld's model was derived (in graphical form only) from clear-sky radiation data measured on vertical planes exclusively. It is not considered here because of these limitations. However, a numerical fit to Threlkeld's data was provided by Stephenson (1965). This was later adopted by the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), which generalized the model by simply specifying the isotropic formula for all tilts different from vertical (ASHRAE, 2005). This composite ASHRAE model is important in practice since it is used routinely by building engineers for solar heat gain and cooling load calculations. In ascending chronological order, the other models considered here have been proposed by Temps and Coulson (1977), Klucher (1979), Hay (1979), Skartveit and Olseth (1986), Gueymard (1987), Perez et al. (1990), Reindl et al. (1990), and Muneer (2004). Note that the Reindl model is also referred to as "HDKR" (Duffie and Beckman, 1991). These models are well described in their original publication, and in the abundant literature on this topic as well, so that no further details are provided for conciseness.

To use Eq. (1), the radiative input must be in the form of separate measurements of the three components E_{bn} , E_d and E , or of only two of them through the additional use of Eq. (2). At many experimental sites, only one irradiance component (usually E) is measured. In the majority of cases in practice the solar resource must be predicted at a non-instrumented site, so that E needs to be interpolated from other sites, or evaluated first through appropriate modeling and meteorological data. The accuracy of this predetermination of E conditions the accuracy of E_s . Since there is a large number of ways to estimate E at any site, the propagation of errors from E to E_s can be extremely difficult to evaluate. Such a specific error analysis has been left out of the scope of this study. Therefore, it is assumed here that at least E is known from on-site measurement.

If only E is known, E_{bn} and E_d must be estimated. To separate the direct and diffuse components from E , many empirical relationships have been proposed in the literature. Most such correlations are univariate (e.g., Erbs et al. 1982; Orgill and Hollands, 1977), i.e., they use global irradiance as the only independent variable.

Among the various versions of the Reindl model, the most detailed approach is multivariate: the direct/diffuse separation is made dependent on Z , ambient temperature, and relative humidity. Between these univariate and multivariate approaches, a bivariate model (Maxwell, 1987) separates DNI and diffuse radiation using a suite of functions of global irradiance and Z . The Erbs, Orgill, Reindl and Maxwell correlations are widely used and considered as relatively "universal". They are representative of the many empirical functions that have been proposed in the abundant literature on this topic, albeit with three perceptibly

different levels of sophistication. They will be used concurrently in what follows to derive E_{bn} and E_d from E .

3. Experimental data

The datasets used here are from NREL's Solar Radiation Research Laboratory in Golden, Colorado (latitude 39.74°N, longitude 105.18°W, elevation 1829 m). This site is located on a mesa that overlooks the western side of the agglomeration of Denver. This is a sunny site, with an annual daily-average DNI of about 5.65 kWh/m². Such a resource can be classified as “relatively favorable” for solar applications. [This average DNI is close to the critical threshold of 6.0 kWh/m², above which the resource is generally considered favorable to the construction of solar power plants (Gueymard et al., 2002).]

The datasets have been obtained from SRRL's download tool, http://www.nrel.gov/midc/srrl_bms. The 1-min irradiance data used here are from the following thermopile instruments: a Kipp & Zonen (KZ) CH1 pyrheliometer to measure DNI, a KZ CM22 to measure diffuse horizontal irradiance (both ventilated and with a tracking shade). The combination of the CH1 and CM22 instruments provides the “optimal” measurement dataset for direct and diffuse radiation. The optimal global irradiance is obtained by summation of these components per Eq. (2), following the current best practice (see, e.g., Gueymard and Myers, 2008b, 2009). This, and all the other steps taken to improve the measurement quality, should guarantee expanded uncertainties in the reference E_{bn} , E_d and E better than 2.0%, 5.0% and 5.4%, respectively.

An Eppley PSP with shadowband, combined with the CH1, defines a suboptimal way to obtain global irradiance. An unshaded PSP measures global horizontal radiation and provides another source of suboptimal data. A KZ CM21 measures E_s on a 2-axis tracking plane always normal to the sun's direction, and five tilted PSPs measure E_s (i) on a south-facing tilt of 40° [nearly equal to latitude], and (ii) on vertical planes that face the four cardinal directions. The uncertainty in the tilted data is ≈3% for 2-axis global and ≈5% for the fixed tilts.

Ancillary data, also measured at 1-min intervals, include: temperature, relative humidity, total cloud cover, opaque cloud cover, ground albedo, and net thermal infrared irradiance at the radiometers. The latter data is used to optimize the diffuse irradiance measurements per the correction method recently proposed (Gueymard and Myers, 2009; Michalsky et al., 2007). The cloud cover information is obtained from a Yankee TSI instrument with a spatial resolution of 1% sky cover. The ground albedo is obtained as the ratio of the upwelling and downwelling global irradiance measured by two PSPs at a height of 1.6 m. The vertical, 40° and 2-axis tracking instruments are all installed on a tower, with platforms at 2.4, 3.4, and 4 m above ground, respectively. This tower is about 52 m away from the setup measuring the horizontal irradiance and DNI components.

The experimental datasets cover the 12-month period September 2006 to August 2007. Their quality has been thoroughly controlled, adding to the normal QC tests routinely performed by NREL. For instance, all data points with missing data from any sensor, with negative short-wave irradiance, with significant differences in redundant measurements, or with albedo values beyond some acceptable limits (0.05–1.0), have been rejected. A particularly insidious problem arises whenever the tilted radiometers sense direct radiation during a longer portion of the 1-min period than the horizontal ones, or vice versa. Error bounds have been used to eliminate these undesirable data points. It is stressed that the simultaneity of data collected by different instruments that are not exactly collocated is the more an issue that the measurement frequency increases. In contrast, conventional hourly data would not be affected as much, if at all. A total of 116,942 data points met all these criteria, and served as the main database for further analysis.

In addition to this “Reference” case, five alternate sub-optimal input datasets have been devised:

- (1) global horizontal calculated from Eq. (2) as before, but where diffuse irradiance is measured with a PSP/shadowband combination (case noted “Shadowband”);
- (2) global horizontal measured with a PSP, and used with the Erbs et al. separation method (case noted “Erbs”);
- (3) global horizontal measured with a PSP, and used with the Orgill & Hollands separation method (case noted “Orgill”);
- (4) global horizontal measured with a PSP, and used with the Maxwell separation method (case noted “Maxwell”);
- (5) global horizontal measured with a PSP, and used with the Reindl et al. separation method (case noted “Reindl”).

These five alternate cases try to reproduce what would be typical in the practice of solar engineering calculations, i.e., use of suboptimal data.

To better evaluate how the results obtained here can be representative of other climatic conditions, the accuracy of the prediction of E_s has also been analyzed as a function of season (summer *vs* winter, conducive to high *vs* low sun), sky clarity (clear-sky *vs* all-sky conditions), snow-free ground (low albedo) *vs* snow on ground (high albedo), and ground albedo estimated at a constant value of 0.2.

4. Transposition models performance

In this section, the *intrinsic* performance of the ten transposition models is evaluated by considering only the reference case where *optimal* input data are used. Each 1-min prediction of E_s via Eq. (1) is compared to its measured counterpart. This defines the individual difference between

prediction and measurement, or *apparent* error of the prediction. It must be stressed that this is only an apparent error since each measured data point also carries some systematic and random error in it. To conform to common usage, the term “error” will be used in what follows, even though the term “difference” would have been more appropriate.

The dataset is analyzed first as a whole, then for various subsets. The usual summary statistics (mean bias error, MBE, and root mean square error, RMSE) are obtained for each of the ten transposition models and various tilted planes. More performance indices do exist, and are detailed elsewhere (Gueymard and Myers, 2008a). However, the latter reference showed that these indices behave differently, so that, with present knowledge, any “objective” model ranking attempt is in fact index-dependent. The assessment described below is therefore not aimed at ranking the ten models, but, more modestly, at evaluating how their prediction uncertainty compares to that of the measurement, and is affected by atmospheric or environmental conditions. Considering that a bias in the tilted pyranometers’ calibration of about $\pm 1\%$ may exist, more weight is given to the random errors in the following discussion.

4.1. Overall statistical results

The summary statistics for all models and three tilt geometries (40°S, 90°S, and fully 2-axis tracking) appear in Table 1. These results are separated into two important

groups, depending on cloud cover: (i) all-sky conditions (any cloud cover); and (ii) clear-sky conditions only (defined here as cloud amount $\leq 10\%$ and DNI ≥ 120 W/m²). The latter case constitutes as much as 50% of all cases, owing to the predominantly clear atmosphere at SRRL.

The results in Table 1 clearly indicate that the isotropic approximation underestimates systematically and is the poorest performer in all cases, which could be expected, considering the overwhelming evidence provided by the literature. The ASHRAE model is therefore affected by the same problem, except for the vertical tilt (since it uses an anisotropic algorithm in that specific case).

The clear-sky conditions (bottom of Table 1) are obviously associated with higher irradiances than those for all-sky conditions, but only 19–22% more on average because of the predominantly clear atmosphere. The RMS errors are lowest under clear-sky conditions, for all models and tilts. This could be expected because (i) the sky diffuse irradiance is low under clear skies, particularly at this clean, high-elevation site, compared to the high direct irradiance (considered “errorless” in the present context); (ii) clear-sky irradiances only change slowly and smoothly over one-minute intervals; and (iii) the sky radiance is spatially much more homogeneous than under partly-cloudy conditions, for instance. The latter two reasons suggest that these 1-minute RMS errors should be representative of the corresponding *hourly* RMS values, therefore easing the process of comparing the present results to others of the literature. The MBEs are below

Table 1

Performance of ten transposition models (in alphabetical order) when using reference input data and the whole 12-month dataset. Percent results refer to the mean plane-of-array irradiance E_s for each plane. Top pane: results for all-sky conditions; bottom pane: results for clear-sky conditions only.

Plane	40°S		90°S		Tracking	
Model	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)
All-sky, $N = 116,927$						
Mean E_s (W/m ²)	643.2		432.3		835.9	
ASHRAE	−5.1	7.8	6.5	13.4	−8.1	9.6
Gueymard	−0.8	4.3	5.2	10.6	−0.9	4.2
Hay	−2.1	5.5	−2.7	8.2	−1.9	6.1
Isotropic	−5.1	7.8	−5.8	11.6	−8.1	9.6
Klucher	−1.4	4.6	0.3	8.5	−6.0	7.5
Muneer	−0.4	5.2	2.5	9.4	−5.4	7.0
Perez	−2.7	6.7	−4.7	12.0	−2.3	5.8
Reindl	−1.8	5.3	−0.4	7.7	−1.5	5.9
Skartveit	−2.4	5.7	−4.3	9.3	−2.2	6.4
Temps	1.3	6.0	4.8	11.3	−4.5	6.6
Clear sky, $N = 58,871$						
Mean E_s (W/m ²)	763.0		523.5		1019.6	
ASHRAE	−3.6	4.9	0.1	4.5	−5.5	6.2
Gueymard	−1.1	2.7	−0.2	3.8	−1.2	2.1
Hay	−1.0	3.2	−2.2	5.2	−0.3	2.7
Isotropic	−3.6	4.9	−4.9	8.1	−5.5	6.2
Klucher	−1.0	2.8	−0.5	5.1	−4.1	4.8
Muneer	0.4	2.8	2.0	5.1	−3.3	4.0
Perez	−0.7	2.6	−0.4	3.9	−1.0	2.1
Reindl	−0.9	3.1	−1.2	4.7	−0.1	2.7
Skartveit	−1.0	3.2	−2.2	5.2	−0.3	2.7
Temps	−0.9	2.7	−0.3	5.0	−4.0	4.7

$\pm 5\%$ in all cases (except with the isotropic assumption), and would be conserved if hourly periods were rather considered. The relatively low bias and random errors obtained with all anisotropic models suggest that their clear-sky predictions are of comparable accuracy to that of the actual measurements. Under such ideal conditions, the Gueymard and Perez models exhibit quite consistently lower bias and random differences, and therefore appear to perform better (particularly in the tracking plane case).

When considering both clear and cloudy conditions (top of Table 1), all models suffer of a noticeable performance drop, indicated by larger MBE and RMSE. The RMS results show that only two models (Gueymard and Klucher) are below the 5% level for the 40°S plane, and only one (Gueymard) for the tracking plane. RMS errors are far larger—and substantially beyond the experimental uncertainty—for the south-facing vertical plane. The Reindl model performs best in the latter case. Since cloudy sky conditions are rapidly changing, it is likely that hourly RMS values would not be as high as the reported 1-minute values. Conversely, random errors would likely be much larger at cloudier sites. Interestingly, some models appear to be affected more than others by the presence of clouds. This translates into variable model-to-model changes in MBE and RMSE between the clear-sky and all-sky cases. This might explain, at least in part, why the relative performance of anisotropic models is location-specific, as the literature reveals. Some models simply happen to be better tuned to very clear conditions, others to mid-range cloudiness, and others to heavy cloudiness. This is also apparent in previous results (Gueymard and Myers, 2009), where the performance structure of various transposition models was found to be dependent on the frequency distribution of the E_d/E ratio, which is a strong function of cloudiness.

The significantly lower performance of all transposition models for the vertical plane can be related to a drastic change in the relative importance of the different components in E_s , per Eq. (1). From Eq. (2), R_r is 0.5 for a vertical tilt, but only 0.117 for $s = 40^\circ$. This means that the prediction of E_s for a vertical plane becomes very sensitive to any inaccuracies in the evaluation of the effective foreground reflectance, including its anisotropic features. Moreover, the mean vertical E_s is also less rich in direct radiation (which is considered “errorless” here)—about 33% less overall than what is incident on the 40° tilt, on average.

Another important aspect of these results is how they may apply to the design of solar systems. In Golden or similar sunny sites, 2-axis tracking flat-panel solar systems receive on average $\approx 30\%$ more radiation than a flat-plate collector tilted at latitude. Under clear skies, this difference increases to $\approx 34\%$, and the mean annual global normal irradiance reaches, and even exceeds, the “one sun” level commonly referred to by energy ratings, i.e., 1000 W/m². Under such highly favorable conditions, the Gueymard and Perez models perform consistently better than the others, under either clear-sky or all-sky conditions.

The anisotropic reflectance issues mentioned earlier (and discussed further in the next section) appear even more obvious when considering the irradiance on vertical tilts of other cardinal directions, also measured at NREL. Although cardinal directions other than South (for the northern hemisphere) are of minor interest in solar energy applications (and are therefore not discussed in detail here), they are still important for the calculation of solar heat gains through windows, for instance. The all-sky performance of all models appears to degrade considerably when the vertical surface’s azimuth changes from S to E, W and N. The concomitant spread in mean annual RMS values changes from 7.7–13.4% (Table 1) to 10.5–20.0%, 13.3–26.9%, and 14.0–54.0%, respectively. It is highly likely that a significant part of these increasingly large apparent errors is in fact due to the presence of experimental artifacts (from shading, parasitic reflections and inhomogeneous conditions) coupled with oversimplified modeling of ground reflection, and with completely ignored obstruction shading in current transposition models. The large error increase in the case of vertical surfaces is consistent with previous results from the literature. Since no known experimental setup can exactly replicate the ideal conditions assumed by transposition models, it is argued that selecting or ranking models according to their apparent performance relative to vertical surfaces should not be attempted unless the sky diffuse and ground-reflected diffuse irradiances can be precisely separated, which is not the case here.

4.2. Season and albedo effects

The 12-month dataset has been split into two ideal seasons, covering the periods April–August (dubbed “Summer”), and September–March (dubbed “Winter”). The “Winter” period was further split into “no snow” and “snow on ground” subsets, using a threshold albedo value of 0.5. No occurrence of snow-covered ground was found in “Summer”. Finally, a fourth dataset was created with all data points of any season by using a fixed albedo of 0.2 rather than the measured albedo. This is the usual simplification whenever the local albedo is not measured, i.e., in the vast majority of cases in an engineering practice.

The interpretation of the summary statistics in Table 2 is difficult because the effects of season (and therefore solar geometry) and ground albedo are not the same for all models or tilts. Interestingly, however, all predictions for the vertical tilt appear more accurate in winter with snow-covered ground, i.e., when both the ground-reflected irradiance and E_s are at their maximum.

The ground albedo varies during the day for various reasons, including departure from Lambert’s Law of isotropy, and changes in ground properties (such as soil’s water content or snow cover). The diurnal variation of the measured albedo is shown in Fig. 1 for two clear summer days and one clear winter day. The latter case shows a rapid decrease in albedo caused by melting

Table 2

Performance of ten transposition models (in alphabetical order) when using subsets of reference data, showing season and albedo effects.

Plane	40°S		90°S		Tracking	
Model	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)
Summer, no snow, $N = 61,669$						
Mean E_s (W/m ²)	614.8		285.7		857.7	
ASHRAE	−3.2	6.1	11.2	18.9	−7.3	8.7
Gueymard	−1.0	4.1	8.4	16.1	−2.0	3.9
Hay	−2.7	4.9	−4.7	9.2	−2.7	5.0
Isotropic	−3.2	6.1	0.1	7.2	−7.3	8.7
Klucher	−0.8	4.7	7.7	11.6	−5.4	6.8
Muneer	−0.6	4.0	3.9	10.8	−5.0	6.4
Perez	−3.5	7.4	−8.7	19.4	−3.0	5.6
Reindl	−2.4	4.6	−1.0	8.8	−2.4	4.8
Skartveit	−3.0	5.0	−7.0	9.8	−2.9	5.2
Temps	0.6	5.3	12.2	17.8	−4.3	6.1
Winter, no snow, $N = 36,454$						
Mean E_s (W/m ²)	645.9		487.4		748.5	
ASHRAE	−8.4	10.2	6.8	12.7	−10.6	12.3
Gueymard	−1.7	4.4	4.7	8.8	−0.6	4.6
Hay	−3.5	6.2	−3.5	8.4	−3.1	7.8
Isotropic	−8.4	10.2	−10.1	13.5	−10.6	12.3
Klucher	−3.0	5.0	−3.8	7.9	−7.8	9.6
Muneer	−2.0	5.8	1.0	9.5	−7.2	9.2
Perez	−3.5	6.5	−4.1	9.9	−2.7	6.8
Reindl	−3.2	5.9	−1.3	7.5	−2.4	7.4
Skartveit	−3.8	6.5	−5.2	10.2	−3.5	8.3
Temps	1.8	6.3	2.3	8.7	−5.4	7.9
Winter, with snow, $N = 18,808$						
Mean E_s (W/m ²)	731.6		806.1		934.1	
ASHRAE	−5.0	6.9	0.7	7.6	−6.7	8.0
Gueymard	1.6	4.4	1.9	6.3	1.6	4.2
Hay	1.9	5.5	0.5	5.9	2.2	6.2
Isotropic	−5.0	6.9	−7.6	9.5	−6.7	8.0
Klucher	−0.6	3.7	−3.6	5.6	−5.2	6.4
Muneer	2.9	6.2	2.5	6.9	−3.9	5.4
Perez	0.7	4.8	−0.9	5.9	0.4	4.7
Reindl	2.0	5.5	1.1	5.9	2.5	6.3
Skartveit	1.7	5.6	−0.2	6.4	1.8	6.4
Temps	2.5	7.0	−0.8	6.8	−4.1	6.0
Albedo = 0.2, $N = 116,927$						
Mean E_s (W/m ²)	643.2		432.3		835.9	
ASHRAE	−5.6	8.3	3.8	18.3	−9.2	11.0
Gueymard	−1.2	4.3	2.4	15.6	−2.0	4.9
Hay	−2.5	5.3	−5.5	13.3	−3.0	6.2
Isotropic	−5.6	8.3	−8.5	19.5	−9.2	11.0
Klucher	−1.9	4.9	−2.4	16.3	−7.1	9.0
Muneer	−0.8	4.9	−0.2	13.6	−6.5	8.3
Perez	−3.2	6.7	−7.4	16.8	−3.4	6.6
Reindl	−2.2	5.1	−3.2	12.9	−2.6	6.0
Skartveit	−2.8	5.5	−7.1	14.1	−3.3	6.6
Temps	0.9	6.0	2.1	17.3	−5.6	8.1

snow. Even the albedo of dry ground varies during the day in summer, with a minimum around noon. The morning and afternoon albedos are not symmetrical, due to azimuthal inhomogeneities in ground cover and possible partial shading. Moreover, the early morning and late evening albedos are often close to either 0 or 1, mostly due to artifacts, such as shading or instrumental cosine error. Over the whole dataset, the winter albedo varies between 0.05 and 1.0 (because of the quality-control constraints mentioned earlier), with a mean of

0.375 ± 0.277 . Similarly, for summer, the minimum and maximum are 0.05 and 0.552, with a mean of 0.184 ± 0.033 . The annual average albedo is 0.274, i.e., somewhat larger than the conventional value of 0.2. As could be expected, errors increase noticeably when assuming a fixed albedo of 0.2 rather than using its measured value (compare the bottom part of Table 2 to the top part of Table 1). As could be expected also, this error increase is far more important for a vertical surface than for more moderate tilts.

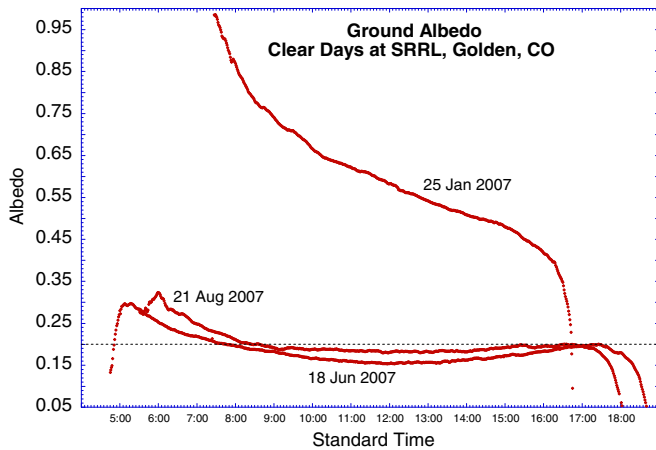


Fig. 1. Diurnal variation of 1-minute ground albedo measured during two summer clear days and a winter day (with snow on ground) at SRRL. The dashed line indicates the conventional constant value of 0.2.

The uncertainty in ground reflection modeling, and of the compensation of errors that results (for some models more than others), is another factor that can explain (at least in part) why models perform differently depending on tilt geometry, or why they also perform differently from one specific site to the other, as the literature reveals. The present results confirm the concerns previously expressed about the importance of correctly measuring or estimating the ground-reflected irradiance (Gueymard, 1987; Ineichen et al., 1987, 1990; Loutzenhiser et al., 2007).

5. Suboptimal data uncertainty

In most practical cases of an engineering practice, the inputs to the transposition models reviewed above are not optimal, either because of the experimental inadequacies summarized in a previous study (Gueymard and Myers, 2009), or because of the uncertainties introduced by the intermediate calculations needed to obtain E , E_{bn} and E_d whenever these irradiances are not measured locally (which is the most frequent case).

In the Shadowband case (where global is obtained as the sum of shadowband diffuse and direct horizontal, as described in Section 3), only the diffuse measurement is suboptimal, resulting in an average RMS error of 7.5% in measured diffuse horizontal and 2.4% in derived global horizontal (Table 3). The reference DNI is used in this case, with therefore no error in what constitutes 68% of the average global horizontal irradiance.

The main input of the direct/diffuse separation methods is the global horizontal irradiance, obtained here from an unventilated and thermally uncorrected Eppley PSP pyranometer. This instrument has a large installed base over the world, so that this setup can be considered typical. The PSP-measured global irradiance yields an RMS error of $\approx 5\%$, indicating a less optimal setup than the Shadowband just mentioned, particularly under clear skies. This might seem contrary to conventional knowledge, but there

are specific reasons for this, which have been discussed previously (see, e.g., Gueymard and Myers, 2009).

As the summary statistics in Table 3 also reveal, all the separation methods add significant random errors in the process, with typical RMS errors in DNI of $\approx 15\%$ under all-sky conditions and 5–9% under clear-sky conditions. The performance of the Erbs method is similar in the all-sky and clear-sky cases. The Orgill method produces results very similar to those of the Erbs method. The Reindl method is less consistent, having the lowest RMS in the former case and the largest in the latter case. Conversely, the Maxwell correlation has the largest biases under all-sky conditions, but becomes the best performer to predict the direct and diffuse components under clear-sky conditions. All methods, except Maxwell's, significantly underestimate DNI and overestimate diffuse under clear-sky conditions, with an opposite trend under all-sky conditions. As discussed elsewhere (Gueymard, 2005), a possible reason for this is the low aerosol turbidity, which is prevailing at this site but not taken into consideration by any such separation method. These findings confirm previous results that have been obtained for other low-turbidity locations of the Southern Hemisphere (Spencer, 1982). The large RMS errors obtained here are consistent with the results from the many studies of the literature, which used hourly data most generally. Most troubling, however, is that the two more elaborate separation methods (Maxwell and, even more so, Reindl) do not perform better than the simpler ones (Erbs and Orgill). To the contrary, there are conditions (season or sky clarity) for which the latter ones yield lower bias or random errors than Maxwell or Reindl. This can be explained by the fact that, while all separation methods are empirical in nature and may not apply as well to all possible climates, the multivariate methods may still ignore the physically most important variables (such as cloud properties and aerosol turbidity).

6. Suboptimal tilted irradiance prediction

The effect on the predicted tilted radiation of combining suboptimal input data and suboptimal direct/diffuse separation is now analyzed. As can be expected, the more their input data deviate from optimal quality, the more prediction errors increase. This is shown in Fig. 2 for the 40°S tilt, for instance. In this case, the incident tilted irradiance is only marginally affected by any modeling error in foreground reflectance (since R_r is small), and the largest step in error increase results from using an empirical direct/diffuse separation instead of measured data. The concomitant error increase is not uniform from one transposition model to the other. It is relatively larger for the Gueymard, Klucher, Muneer and Temps models than for the Hay, Perez, Reindl and Skartveit models when using the Erbs separation method (which overestimates the direct/diffuse ratio), for instance. Similar results are obtained with the Orgill separation method. Conversely, different compensations of errors appear to occur when using either the Maxwell

Table 3
Mean annual errors in horizontal irradiance due to using suboptimal measurement and estimation techniques.

Component	Global		Diffuse		Direct	
Model	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)
All-sky, $N = 116,927$						
Reference (W/m^2)	543.5		173.8		598.2	
Ref. DNI + Shadowband Diff.	−0.1	2.4	−0.2	7.5	0.0	0.0
Meas. Global + Orgill	−0.4	4.5	−8.2	46.9	2.1	14.9
Meas. Global + Erbs	−0.4	4.5	−10.6	47.4	2.8	15.1
Meas. Global + Maxwell	−0.4	4.5	−24.2	48.2	6.7	14.9
Meas. Global + Reindl	−0.4	4.5	1.1	46.6	−0.6	14.2
Clear sky, $N = 58,871$						
Reference (W/m^2)	620.5		90.7		855.1	
Ref. DNI + Shadowband Diff.	0.3	1.7	2.0	11.5	0.0	0.0
Meas. Global + Orgill	−0.9	4.7	39.6	55.4	−4.9	7.1
Meas. Global + Erbs	−0.9	4.7	32.3	49.5	−4.1	6.7
Meas. Global + Maxwell	−0.9	4.7	−8.6	39.6	0.2	5.2
Meas. Global + Reindl	−0.9	4.7	57.8	82.7	−6.8	9.1

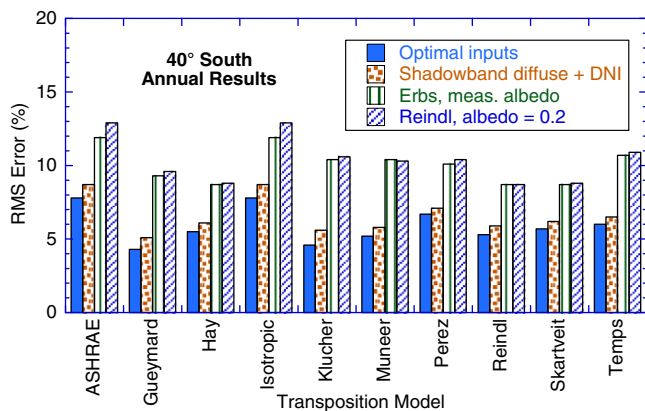


Fig. 2. Apparent RMS error for ten transposition models for 40° tilt facing south at SRRL. Four input data quatties are compared: (i) Reference (optimal) measurements of direct and diffuse irradiance and of albedo; (ii) Reference direct irradiance measurement, suboptimal diffuse irradiance measurement with shadowband, and measured albedo; (iii) Measured global irradiance with direct/diffuse separation using the Erbs correlation, and measured albedo; and (iv) Measured global irradiance with direct/diffuse separation using the Reindl correlation, and fixed albedo value of 0.2.

separation method (which overestimates the direct/diffuse ratio even more) or the Reindl separation method (which is relatively neutral for all-sky conditions on a yearly basis). A better understanding of these error compensations is desirable so that some generalization can be eventually obtained and “universal” recommendations can be proposed. A thorough investigation comparing the frequency distributions between model errors and direct/diffuse or diffuse/global ratio at various sites is suggested to that effect, but is beyond the scope of this contribution.

For conciseness, only a subset of annual results from all possible combinations is compiled in Table 4. Moreover, the results for the Orgill case are omitted because very similar to those for Erbs. For all cases, a comparison between Tables 1 and 4 clearly confirms the results shown in Fig. 2: the random error in the predicted E_s increases sharply when replacing optimal input data by suboptimal data.

For all tilts, most separation and transposition model combinations tend to underestimate E_s . This is particularly evident in the case of the vertical plane (90°S) when a fixed albedo of 0.2 is used. In that case, the RMS error also becomes large (Fig. 3). This could be expected since a vertical plane receives a relatively large fraction of ground-reflected radiation, and a fixed value of 0.2 grossly underestimates the albedo in the presence of snow. From an engineering standpoint, such an underestimation may be a good thing (to some extent), since it is on the safe or conservative side. The problem is that, *a priori*, it is difficult to know whether an underestimation or an overestimation will occur at any specific site and/or for any specific period.

More generally, it is nearly impossible to make recommendations about the best model combination because of the number of variables involved, the various possibilities of error cancelation, and the relative importance of bias (MBE) *vs* random errors (RMSE) in the decision process. The main finding of this study, which needs to be emphasized, is that in most cases, the intrinsic uncertainty of a transposition model becomes secondary when combined with suboptimal global radiation data, a direct/diffuse separation model, and a rough estimate of the ground albedo.

Fig. 4 compares the performance results (expressed as percent RMS errors) of the ten transposition models for different seasons/conditions, the 2-axis tracking plane, and the ideal case where all inputs are optimally measured. These results can be directly compared to those of Fig. 5, rather obtained for the pragmatic case where the Erbs correlation does the direct/diffuse separation in the input data. The degradation in performance between Figs. 4 and 5 is obvious and significant. Moreover, the model performance ranking that can be attempted from the results in Fig. 4 does not translate necessarily well to the results in Fig. 5. The only possible exception is the isotropic transposition model, which consistently appears as the worst performer.

The selection of the most appropriate combination of models and albedo value ultimately conditions the performance of the whole chain of calculations. Therefore, it

Table 4

Performance of ten transposition models when using suboptimal data obtained from three empirical direct/diffuse separation techniques.

Plane	40°S		90°S		Tracking	
Model	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)	MBE (%)	RMSE (%)
Mean E_z (W/m ²)	643.2		432.3		835.9	
All-sky, Erbs correlation, $N = 116,927$						
ASHRAE	−4.8	11.9	4.3	20.1	−8.5	16.4
Gueymard	−0.8	9.3	3.2	17.5	−1.6	12.1
Hay	−2.0	8.7	−3.6	14.9	−2.3	11.3
Isotropic	−4.8	11.9	−6.5	20.3	−8.5	16.4
Klucher	−1.0	10.4	−0.1	18.8	−6.3	15.5
Muneer	−0.1	10.4	1.6	18.3	−5.8	15.3
Perez	−1.2	10.1	−2.2	17.8	−1.7	13.3
Reindl	−1.6	8.7	−1.2	14.8	−1.9	11.2
Skartveit	−2.2	8.7	−5.0	15.0	−2.6	11.3
Temps	0.8	10.7	2.9	19.3	−5.3	15.2
All-sky, Maxwell correlation, $N = 116,927$						
ASHRAE	−3.7	10.2	4.3	19.1	−4.4	12.4
Gueymard	−0.4	8.8	3.2	16.9	0.8	10.7
Hay	−1.3	8.2	−2.5	14.1	0.2	9.9
Isotropic	−3.7	10.2	−5.0	17.6	−4.4	12.4
Klucher	−0.2	9.4	0.7	16.8	−2.5	11.8
Muneer	0.5	9.5	2.3	17.1	−2.1	11.7
Perez	−0.5	9.8	−1.2	17.7	1.3	11.9
Reindl	−1.0	8.2	−0.2	14.4	0.7	10.0
Skartveit	−1.5	8.2	−3.8	14.1	0.0	9.8
Temps	1.1	10.0	2.9	17.9	−1.8	11.8
All-sky, Reindl correlation, $N = 116,927$						
ASHRAE	−5.8	12.1	5.2	20.6	−8.7	15.5
Gueymard	−1.2	9.1	3.6	17.5	−1.5	11.3
Hay	−2.3	8.4	−3.7	14.2	−2.0	10.2
Isotropic	−5.8	12.1	−7.0	20.7	−8.7	15.5
Klucher	−1.3	10.0	0.5	18.9	−6.2	14.2
Muneer	−0.1	9.9	3.1	18.3	−5.5	13.8
Perez	−2.2	9.9	−3.5	17.4	−1.9	12.3
Reindl	−1.9	8.3	−1.0	14.2	−1.5	10.2
Skartveit	−2.5	8.4	−5.0	14.3	−2.2	10.2
Temps	0.5	10.4	3.5	19.7	−5.2	14.0
All-sky, Erbs correlation, Albedo = 0.2, $N = 116,927$						
ASHRAE	−5.2	12.5	1.8	24.9	−9.5	17.6
Gueymard	−1.2	9.7	0.7	22.2	−2.6	12.9
Hay	−2.4	9.0	−6.0	19.6	−3.3	11.9
Isotropic	−5.2	12.5	−8.9	26.5	−9.5	17.6
Klucher	−1.4	10.8	−2.5	24.4	−7.3	16.6
Muneer	−0.5	10.6	−0.9	22.2	−6.8	16.3
Perez	−1.6	10.4	−4.6	22.3	−2.7	14.0
Reindl	−2.0	9.0	−3.6	19.5	−2.9	11.8
Skartveit	−2.6	9.1	−7.4	19.8	−3.6	12.0
Temps	0.4	11.0	0.5	24.5	−6.3	16.6

seems that, from a practical engineering standpoint, the current literature does not provide all the answers needed. This is because, in practice, models cannot be used in the ideal way they have been developed.

7. Conclusion

To the difference of most previous studies on this topic, which have focused on hourly data, 1-minute data have been used here throughout. Usage of these higher-frequency data reflects the current needs from sophisticated simulations of energy production by large-size solar

power plants. The bias errors reported here can be directly compared to previous results based on hourly data because bias is not sensitive to the aggregation time step. Hourly random errors would normally be lower than the 1-minute results obtained here. Under stable conditions, however, the difference between 1-minute and hourly random errors should be small. Although the transposition models reviewed here were generally developed from hourly data, they appear to be equally applicable to 1-minute data. This also means that the main findings presented here should remain valid when considering hourly time steps.

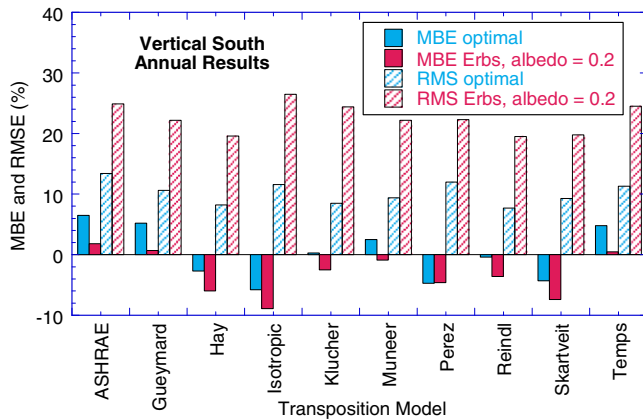


Fig. 3. Apparent Mean Bias and RMS errors for ten transposition models for south vertical tilt at SRRL. Inputs to the models are either reference measured direct and diffuse irradiances using optimal experimental techniques, or measured global irradiance with fixed albedo value and direct/diffuse separation using the Erbs correlation.

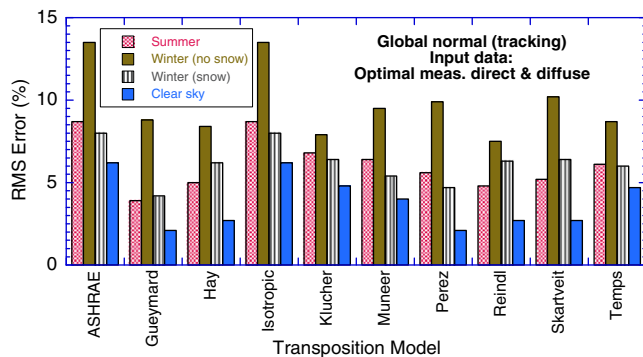


Fig. 4. Dependence of the RMS error for ten transposition models on season and sky clarity for 2-axis tracking plane. Inputs to the models are reference measured direct and diffuse irradiances using optimal experimental techniques.

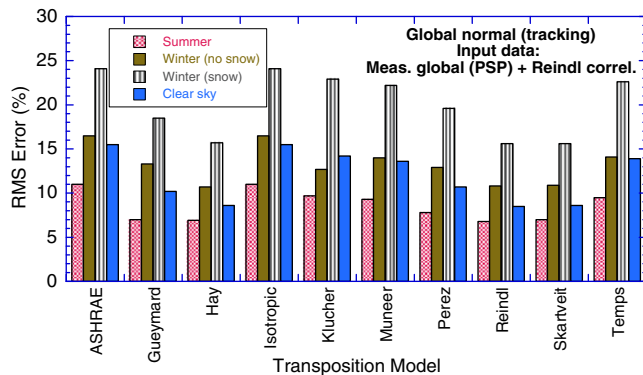


Fig. 5. Dependence of the RMS error for ten transposition models on season and sky clarity for 2-axis tracking plane. Inputs to the models are measured global horizontal irradiance and estimated direct and diffuse irradiances estimated from the Reindl correlation.

It is found that the isotropic transposition method always performs poorly at the low-turbidity SRRL site, confirming previous results of the literature for other sites in widely different climatic areas. In sharp contrast, the per-

formance of all the anisotropic transposition models reviewed here is within, or close to, the instrumental uncertainty limits ($\approx 3\%$ for a 2-axis tracking plane and $\approx 5\%$ for fixed-tilt planes) if optimal input data of measured irradiance (direct + diffuse) and ground albedo are used, and only clear skies are considered. Under such ideal conditions, the Gueymard and Perez models provide estimates of 1-minute global tilted irradiance with low bias and the lowest random differences, compared to global irradiance measured on a 40°S , 90°S and tracking plane. Under all-sky (clear and cloudy) conditions and still ideal input conditions, the performance of all models degrades somewhat, as could be expected. This performance degradation manifests as an increase in both bias and random errors. This is most particularly important for the vertical plane facing south (for which case the Reindl algorithm performs best), and even more pronounced for vertical planes facing the three other cardinal directions. The proposed explanation is that the existing obstacles and inhomogeneous ground conditions in the experimental setup interfere with the ideally perfect environment assumed by all transposition models.

When using suboptimal inputs, i.e., when only global horizontal irradiance is measured locally and the direct and diffuse components need to be estimated, it is found that the performance of all transposition models degrades significantly. Their bias and random error summary statistics vary as a function of receiver geometry, atmospheric conditions, season, ground albedo, and accuracy of the direct/diffuse separation. The four direct/diffuse separation methods that are investigated here behave differently, with noticeable bias and random errors in most cases. The multivariate separation methods (Maxwell and Reindl) do not perform better than the simpler univariate methods (Erbs and Orgill). The transposition models that perform best under these suboptimal conditions are not those that perform best under the optimal conditions mentioned above, due to intricate model-to-model variance in accumulation or compensation of errors. For the site under scrutiny and the various suboptimal input datasets considered, the models performing best are generally those of Hay, Reindl and Skartveit.

The results in this study also show, beyond any doubt, that the major part of the uncertainty in the predicted tilted irradiance at a sunny site is generally caused by the direct/diffuse separation obtained by empirical models whenever these components are not measured locally. For vertical planes and other situations where the ground-reflected radiation is a significant part of the total tilted irradiance, the accuracy of the estimated ground-reflected irradiance becomes another key factor. Consequently, performance assessment results based on first-class experimental measurements and published in the abundant literature on this topic may not apply to the majority of engineering applications in practice, since then the irradiance components and other environmental factors are normally not measured on-site and must be estimated beforehand.

The results presented here are limited to a sunny site where diffuse radiation is lower than average, particularly under clear skies, due to the high altitude and overall clarity of the atmosphere, but should be representative of many other sunny sites with high potential for solar energy applications. Under such circumstances, the inherent inadequacies of the transposition models translate into only limited inaccuracies in the predicted tilted irradiance. What actually limits their performance most is the quality of their inputs, such as the direct and diffuse irradiances or the ground albedo. It is highly likely that a far larger spread in the performance of these models would be found at cloudier sites, in which case the site-specific selection of the most appropriate transposition model might become a more critical issue than here.

Finally, it is recommended that research be now oriented toward the improvement of the complete procedure to predict tilted irradiances in practice, considering the suite of necessary models as a whole, with particular emphasis on reducing the current uncertainties in estimating global, direct and diffuse horizontal irradiances. Better characterization of the ground reflectance processes, including realistic shading of the horizon, is also recommended to improve the prediction of the incident irradiance on steep or vertical surfaces.

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